

Semantic Overlap in Music Tags: A CLAP Embedding Analysis of MTG-Jamendo

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Abstract

Music tags (e.g., genre, instrument, mood) are central to Music Information Retrieval (MIR) and are widely used for search and recommendation, yet their meanings can be acoustically ambiguous: a single tag can sometimes describe multiple distinct kinds of sound. In this work, we study whether tags align with coherent regions in a music embedding space by extracting 512-D CLAP audio embeddings for approximately $[N \approx 55.6k]$ MTG-Jamendo tracks, reducing dimensionality with PCA ($[d=44]$, 85% variance), and fitting Gaussian mixture models to capture non-spherical structure. Using model selection over component count and covariance type, we find that a full-covariance GMM with $[k=15]$ components best explains the embedding distribution, yielding overlapping but interpretable regions when visualized with UMAP. We then quantify ‘tag polysemy’ by measuring the dispersion of each tag over mixture components (entropy and top-1 mass), revealing a spectrum from relatively tight localized tags (e.g., ‘genre-rap’, ‘genre-metal’) to highly dispersed tags (e.g., ‘mood/theme-melodic’, ‘instrument-synthesizer’). These results suggest that tag ambiguity is not merely annotation noise, but reflects systematic multi-modal structure in audio representations, with implications for evaluation and design of tag-based MIR systems.

1. Introduction

Music tags sit at the core of many music information retrieval (MIR) systems, providing useful acoustic descriptors like genre, instrumentation, and mood. They fuel search (“*show me punk rock*”), recommendation (“*more like this upbeat track¹*”), playlist creation, and even how listeners learn to describe music. At the same time, tags are an imperfect interface between language and sound. In

¹ “Track”: industry terminology for “song.” They are used interchangeably in this paper.

practice, the same tag can be applied to songs with noticeably different acoustic characteristics. For example, a melodic rap song might have ‘genre-rap’ and ‘mood/theme-melodic’ tags whereas a melodic folk song may have ‘genre-folk’ and ‘mood/theme-melodic’ tags. “Melodic” can describe many distinct acoustic regions while still accurately describing each of them. Furthermore, a tag might refer to multiple substyles/subgenres, production aesthetics, or musical roles; conversely, two tracks that sound similar may receive different tags depending on context and annotator interpretation. This gap between words and audio motivates our research question: do music tags concentrate in regions of audio embedding space, or do they spread across multiple distinct acoustic modes?

This work studies that question through the lens of modern audio embeddings. Recent representation learning models map each track to a point in a high-dimensional vector space where distances are intended to reflect perceptual, semantic, or acoustic similarity[7]. If such an embedding space meaningfully organizes audio content, then tags should approximately align with regions of that space. But if a tag is acoustically ambiguous—what we can call *tag polysemy*—then tracks sharing that tag may occupy multiple separated regions. Put differently, tag polysemy is not just noise in labeling; it is a structured phenomenon in which a single label corresponds to multiple distinct modes of audio. To clarify the term, ‘polysemy’ is a linguistic term that denotes the coexistence of multiple meanings for a single word or phrase.² It is important to identify and measure this structure because polysemous tags can make it difficult to evaluate auto-tagging systems, make GMM component visualizations harder to understand, and create confusion in tag-based discovery experiences like search, recommendation, and playlist creation.

To investigate tag polysemy at scale, we analyze the MTG-Jamendo [3] music collection using CLAP (Contrastive Language–Audio Pretraining) embeddings [8]. We begin with 55,609 tracks with tag information after filtering. Each track is represented by a 512-dimensional CLAP audio embedding. Because clustering and density modeling can be unstable in very high dimensions, we apply principal component analysis (PCA) to reduce the embeddings to 44 dimensions, retaining

²<https://www.dictionary.com/browse/polysemy>

85.39% cumulative explained variance. We then fit a Gaussian mixture model (GMM) to shape the distribution of songs in embedding space and obtain component (Gaussian cluster) assignments that can be used as a coarse notion of “acoustic regions.” In practice, mixture modeling provides a better fit than spherical clustering, suggesting the embedding space contains non-spherical, overlapping regions (in our experiments on MTG-Jamendo CLAP embeddings, K-means produced weak separation). Finally, we measure how each tag distributes across components using dispersion statistics. Entropy (how spread a tag is across components) and top-1 mass (how concentrated it is in its most common component) were mainly analyzed. Under this framework, a tag that concentrates in one component is localized, while a tag spread across several components is dispersed. In other words, tags that have higher entropy (more spread) and lower top-1 mass (less concentrated) are classified as ‘polysemous,’ and tags with lower entropy (less spread) and higher top-1 mass (more concentrated) are classified as ‘monosemous.’

This approach yields a practical way to connect qualitative intuitions to quantitative evidence (i.e., “*this tag feels broad*” → “*this tag consistently spans multiple acoustic regions*”). For example, ‘genre-hardrock’ appears more localized to Component 7, tending to concentrate in a particular region of embedding space; while ‘genre-psychedelic’ is comparatively dispersed, spreading across multiple regions according to the entropy-based polysemy statistics (see Figure 11). These two examples illustrate the spectrum of tag behavior that motivates this paper: some tags align tightly with a specific acoustic pattern, while others behave more like umbrella terms.

This paper makes three contributions. First, it presents a pipeline for mapping a large tagged music dataset (MTG-Jamendo, specifically) into a learned embedding space and modeling its global structure, including a reproducible path from audio to embeddings to dimensionality reduction and mixture modeling. Second, it introduces a simple quantification of tag polysemy based on how tags distribute over mixture components, allowing tags to be ranked from localized to dispersed. Third, it grounds these statistics in qualitative visualizations and examples that show how different tag categories behave in practice.

2. Problem Background

Music tags, such as genre (‘rock’), instrument (‘guitar’), and mood/theme (‘energetic’) are a central interface between people and large music collections. They support search, recommendation, playlist creation, and browsing, precisely because they are compact and human-interpretable. At the same time, tags are inherently messy: they are produced by people (or systems trained on people) and they often blur together multiple musical ideas. Lamere’s [6] survey of social tagging in MIR emphasizes this duality: tags can encode valuable contextual knowledge (e.g., genre, mood, instrumentation, quality), but they also contain noise, bias, and inconsistency that makes them difficult to interpret as “ground truth.”

A useful lens for this ambiguity is tag polysemy: a tag is polysemous if it corresponds not to one coherent sonic mode, but to a family of distinct acoustic patterns that share a label. This matters because polysemous tags can distort evaluation, reduce interpretability of tag-conditioned retrieval, and complicate attempts to map music collections using embeddings. The goal of this project is therefore not to propose a new tagging model, but to measure how tags behave geometrically: do tags concentrate in a single region of an audio embedding space, or do they systematically spread across multiple acoustic regions?

2.1. MTG-Jamendo: A Benchmark for Tagged Music

To study tag behavior at scale, this work utilizes the MTG-Jamendo dataset [3] created by Bogdanov et al., an open dataset constructed from Jamendo tracks released under Creative Commons licenses and labeled with a fixed vocabulary of 195 human-annotated tags spanning genre, instrument, and mood/theme. The MTG-Jamendo paper’s primary contribution is to provide a standardized dataset, splits, and baselines for automatic music tagging, making it an attractive benchmark.

While MTG-Jamendo provides an essential infrastructure for tagged music, the project mostly treats tags as targets to predict. It does not ask whether a tag label corresponds to one acoustic concept or multiple distinct ones. In this paper, MTG-Jamendo provides a solid dataset launchpad with the consistent tag vocabulary and scale needed to assess tag structure in a meaningful way.

2.2. MWE: Representation Learning for Tagging and Retrieval

MIR literature treats tagging as multi-label prediction and retrieval as ranking, typically optimizing accuracy or retrieval metrics. More recent work explicitly incorporates language representations. For instance, in a paper on Musical Word Embeddings (MWE) [5], Doh et al. argue that generic word embeddings may miss music-specific meaning. They propose learning word embeddings from music-related text and integrating them into an audio–word joint embedding framework for tagging and retrieval.

While MWE primarily aims to improve system performance (better tagging and retrieval), it provides us with inspiration to map tagged music into a joint audio-text space that contextualizes songs with paired tags.

2.3. CLAP: Dual-Encoded Embeddings

CLAP (Contrastive Language–Audio Pretraining) [8] trains audio and text encoders using contrastive learning so that matched audio–text pairs are close in embedding space. The CLAP paper describes design choices and demonstrates strong results on retrieval and classification benchmarks. Figure 1 shows the CLAP model architecture, consisting of an audio encoder and LLM text encoder, which project log-Mel spectrograms for audio and natural language captions for text into high-dimensional embeddings. Then, both are mapped to a shared embedding space. Because CLAP embeddings are shaped by language supervision, they are especially relevant for studying tags: tags are language, and CLAP is designed to align audio with language. This makes CLAP a plausible space in which tag-related structure might be visible, even when downstream analysis uses only audio embeddings.

This paper attempts to bridge the gap between these works by (a) modeling the embedding distribution with a probabilistic model (GMM) in an audio-text space, and (b) defining tag polysemy operationally as dispersion across learned acoustic regions to investigate the behavior of music tags.

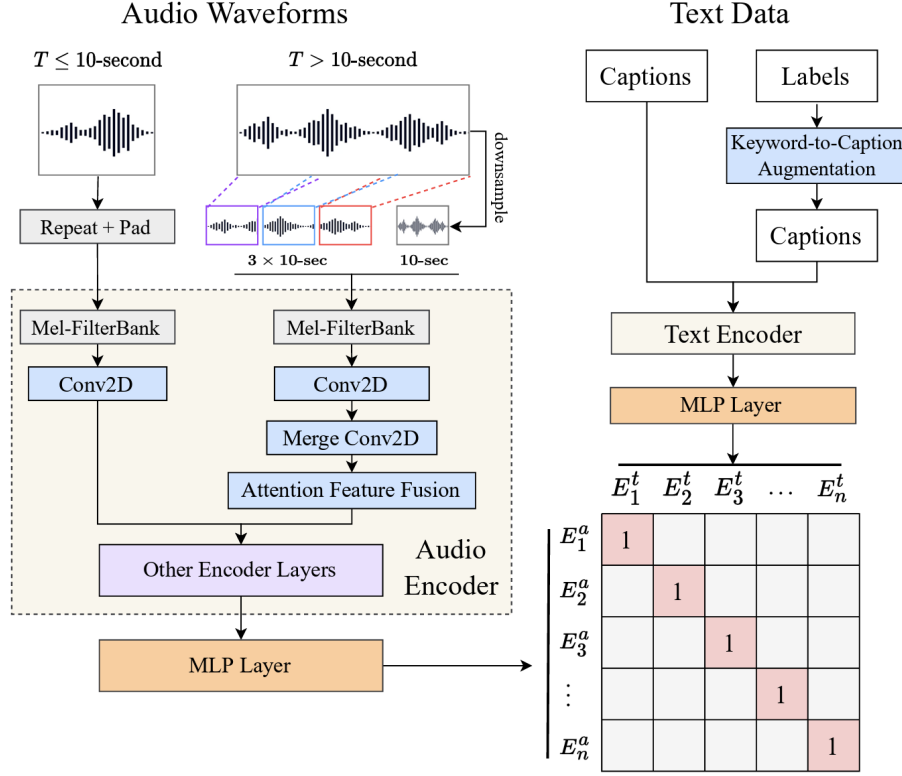


Figure 1: CLAP architecture. [8]

3. Approach and Implementation

3.1. Overview

Our goal is to test whether common music tags (genre, instrument, mood/theme) correspond to coherent regions in a learned audio representation space, or whether they systematically spread across multiple distinct acoustic modes (tag polysemy). We treat the embedding space as a geometric representation of audio similarity and ask: does a tag occupy one dominant region, or several?

To study this at scale, we build a pipeline (see Figure 2) that:

1. downloads MTG-Jamendo audio and tag metadata,
2. maps each track to a high-dimensional audio embedding,
3. models the structure of the embedding distribution using unsupervised clustering/density modeling,

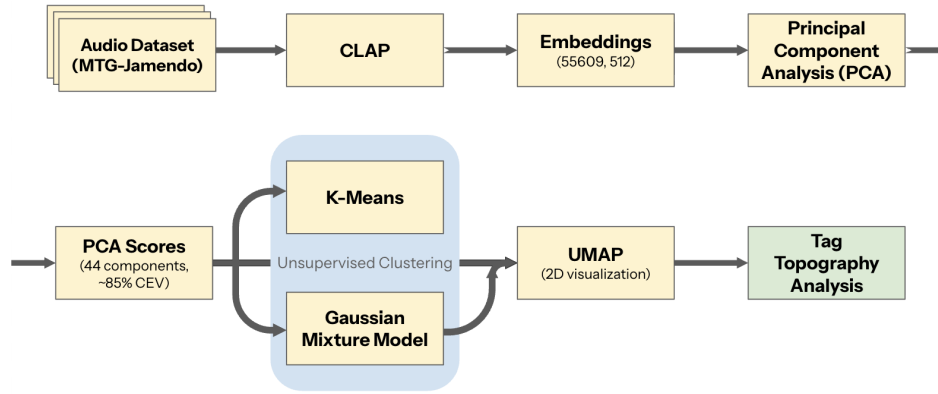


Figure 2: Full project pipeline

4. quantifies each tag's dispersion across the learned components as an operational measure of polysemy, and
5. visualizes each tag's dispersion as a 2D map.

3.2. Environment and Data Storage

All experiments were conducted in Python notebooks (Google Colab) with audio and the codebase³ stored on Google Drive. The MTG-Jamendo audio subset (lower-quality) used in this work totals roughly 156GB, requiring batch-based embedding extraction.

The pipeline is structured across several notebooks for clarity and reproducibility (data retrieval, embedding extraction, EDA, and unsupervised clustering), as seen in Figure 3.

³https://drive.google.com/file/d/1tkYnyFdDLxWJM3qBhnYgWJPYVqYwnqtc/view?usp=drive_link

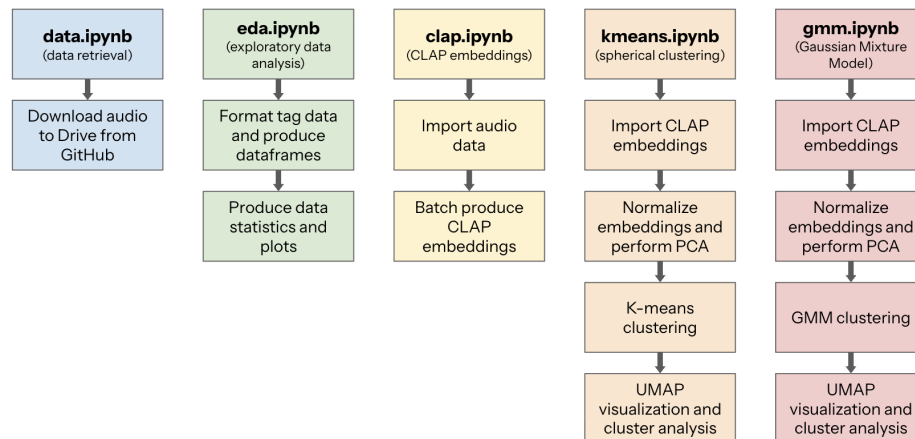


Figure 3: Chart of each notebook and their functions

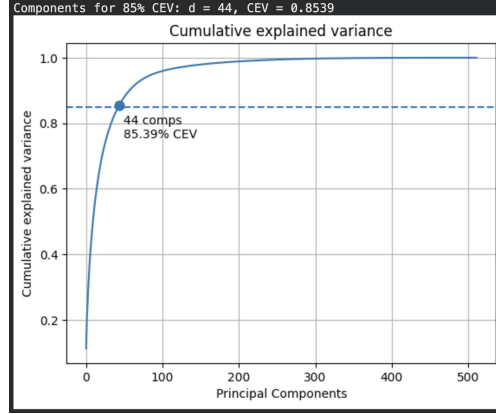


Figure 4: PCA: CEV over Principal Components

3.3. Audio Embeddings: CLAP

In *clap.ipynb*, we first load a checkpoint of the model specifically trained on music.⁴ After loading the dataset from Google Drive, the audio is sent through the CLAP model using the CLAP API[1] and embeddings are saved in batch parquets. After saving and unpacking all batches, the full length of embeddings are accessible in the project folder. Each track i is represented by a 512-dimensional CLAP audio embedding $e_i \in \mathbb{R}^{512}$.

Although CLAP is multimodal, we use the audio encoder only and treat CLAP as an audio feature extractor. Because CLAP is trained to align audio with language, it creates a natural space in which tag-related structure might be visible, even when we do not explicitly use the text encoder during analysis. This isolates the question of whether tag structure is present in audio-derived representations, independent of any explicit text-side modeling.

3.4. Dimensionality Reduction with PCA

Clustering and covariance-based modeling can become unstable and computationally expensive in very high dimensions[2], also known as the “curse of dimensionality.” Therefore, we L2-normalize the CLAP embeddings and apply Principal Component Analysis (PCA), producing PCA scores $d = 44$, which retains 85.39% cumulative explained variance (see Fig. 4), balancing accuracy with efficiency.

⁴music_audioset_epoch_15_esc_90.14.pt’ from MTG-Jamendo GitHub.

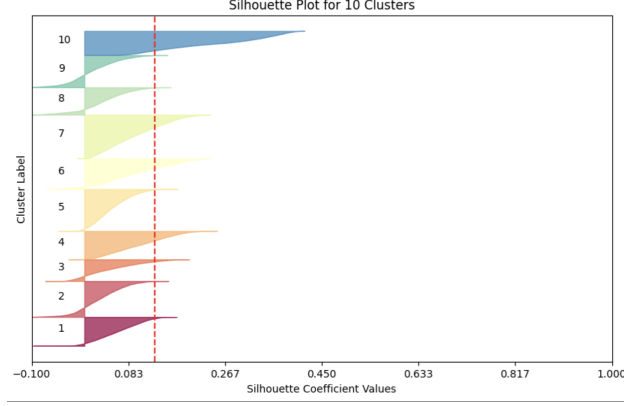


Figure 5: Silhouette Plot

3.5. Baseline: K-means Clustering

As a first attempt to capture structure, in *kmeans.ipynb* we apply K-means clustering to PCA scores and sweep the number of clusters C over a range (approximately 1–30). We compute inertia (within-cluster Sum-of-Squares) and silhouette scores, with an average score of 0.133. Empirically, the inertia plot does not show a strong “elbow,” and silhouette scores remain low with many negative values (see Fig. 5), indicating weak separation and violations of K-means’ spherical-cluster assumption. These observations motivate a more flexible probabilistic model.

3.6. Probabilistic Structure: Gaussian Mixture Model

We model the PCA distribution as a Gaussian Mixture in *gmm.ipynb*:

$$p(\mathbf{x}) = \sum_{c=1}^C \pi_c \mathcal{N}(\mathbf{x} \mid \mu_c, \Sigma_c)$$

where each component c has a mixing weight π_c , mean vector μ_c , and covariance matrix Σ_c . $\mathcal{N}(\mathbf{x} \mid \mu_c, \Sigma_c)$ represents the probability density of data point x under the k -th Gaussian distribution.

Unlike K-means, a GMM can represent ellipsoidal components with different orientations and sizes, and it assigns each track soft membership probabilities rather than a single hard label.

For model selection, we perform a grid search over:

- $C \in \{1, \dots, 31\}$

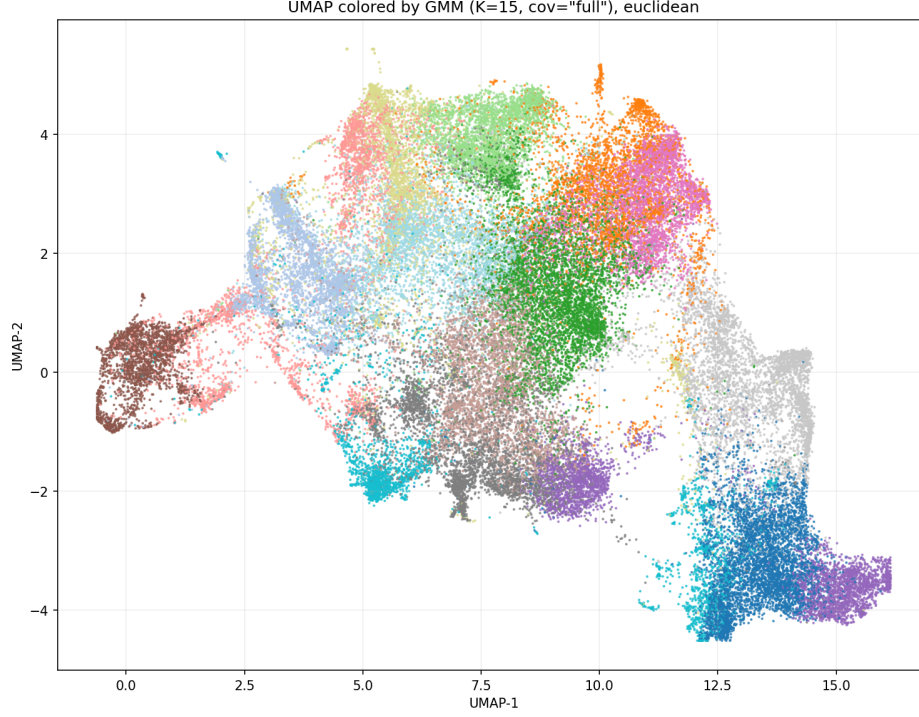


Figure 6: 2D embedding space with UMAP, colored by GMM component.

- covariance types $\in \{spherical, tied, diagonal, full\}$

For each model we compute AIC and BIC, which trade off goodness-of-fit and complexity (lower is better)[4]. The sweep indicates that *full* covariance models consistently achieve the lowest scores, and we select a full-covariance GMM with $C = 15$ as the final model (see Section 4.1).

3.7. Visualization: Uniform Manifold Approximation and Projection (UMAP)

To visualize the learned structure, we apply UMAP to the 44-D PCA scores (in *gmm.ipynb*) to obtain 2D coordinates (u_{i1}, u_{i2}) . We emphasize that UMAP is used for visualization only: it approximately preserves local neighborhoods but is not a precise projection of global geometry. For legibility, the displayed range is clipped to the 1st–99th percentiles to prevent extreme outliers from skewing the axis limits. Points are colored by their GMM component label to reveal distinct but overlapping “islands.” Figure 6 shows the full colored UMAP output.

3.8. Tag Dispersion and Polysemy Statistics

To quantify how consistently a tag maps onto the learned structure of the embedding space, we measure how each tag distributes across component assignments (in *gmm.ipynb*). For each tag t , we collect the set of tagged tracks and compute counts per component. We then turn these counts into a probability distribution over components:

$$p(c | t) = \frac{\text{count}(t, c)}{\sum_{c'} \text{count}(t, c')}$$

Intuitively, $p(c | t)$ measures the fraction of tracks with tag t that fall into component c . Using this distribution, we compute two complementary dispersion statistics. First, we compute entropy⁵ (with natural log):

$$H(t) = - \sum_{c=1}^K p(c | t) \log p(c | t),$$

where larger values indicate that tag t is spread across many components. Because tags may appear in only a subset of components, we also report a normalized entropy,

$$H_{\text{norm}}(t) = \begin{cases} \frac{H(t)}{\log m(t)} & \text{if } m(t) > 1, \\ 0 & \text{if } m(t) = 1. \end{cases}$$

where $m(t)$ is the number of components with a non-zero count for tag t . If tag t only lives in one component, then its entropy is zero. Lastly, we compute top-1 mass:

$$\text{top1}(t) = \max_c p(c | t),$$

which measures how concentrated tag t is in its most common component. Under these definitions, tags with high entropy / low top-1 mass are more dispersed while tags with low entropy / high top-1 mass are more localized.

⁵[https://en.wikipedia.org/wiki/Entropy_\(information_theory\)](https://en.wikipedia.org/wiki/Entropy_(information_theory))

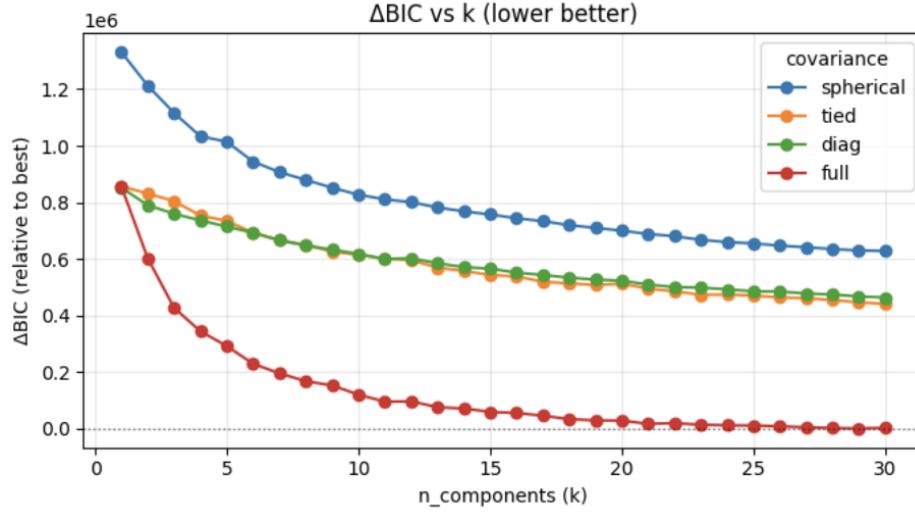


Figure 7: Δ BIC across GMM model sweep.

4. Evaluation

4.1. GMM Model Selection

We fit GMMs over PCA-reduced embeddings across a sweep of component counts $C \in 1, \dots, 30$ and covariance parameterizations $\{spherical, tied, diagonal, full\}$. We compute Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) for each configuration.

Across the sweep, the lowest-scoring models consistently use full covariance, suggesting that the embedding distribution contains non-spherical clusters (i.e., ellipsoidal components) that simpler covariance assumptions fail to capture. In the sweep output, the global minima occurred at relatively large C : the best AIC model used 30 components with full covariance (AIC = -6,183,636.64), and the best BIC model used 26 components with full covariance (BIC = -5,914,470.45). Figure 7 shows AIC/BIC as a function of C and covariance type. It is worth noting that we primarily used BIC for evaluation as it favors simpler models by penalizing model complexity. Rather than using the absolute BIC minimizer, we proceed with a “frugal” full-covariance GMM with $C = 15$ components for downstream analysis (AIC = -5,970,912.13; BIC = -5,832,317.69) since BIC continues improving slowly at larger C while components become harder to interpret and returns diminish.

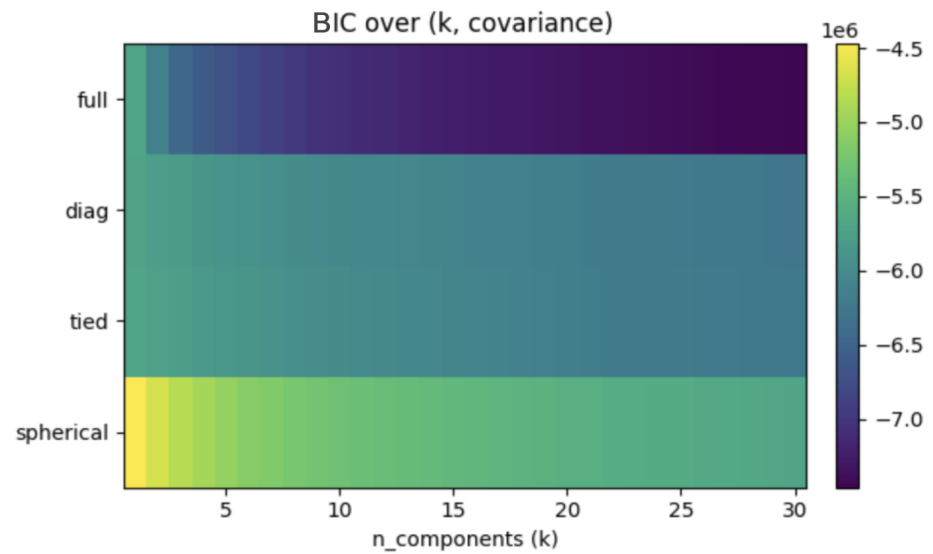


Figure 8: BIC Heatmap. Shows trends in BIC curves in Fig. 7

4.2. Visual Validation with UMAP

To visualize global structure, we project the 44-D PCA space into 2D using UMAP and color points by their hard component label from the $C = 15$, full-covariance GMM. The resulting map shows multiple partially overlapping “islands,” consistent with a multi-modal embedding distribution rather than a single continuous manifold. See Figure 6.

We further validate interpretability with tag highlight overlays, plotting all tracks in gray and highlighting tracks with a given tag (see Figures 9 and 10). Each “island” is numbered and has an approximate ellipse drawn around it to visually distinguish components. These overlays reveal that some tags (e.g., ‘genre-rock’ in Fig. 9) have most of their tracks concentrated in one or a small number of regions, while others (e.g., ‘genre-electronic’ in Fig. 10) appear distributed across many or all regions.

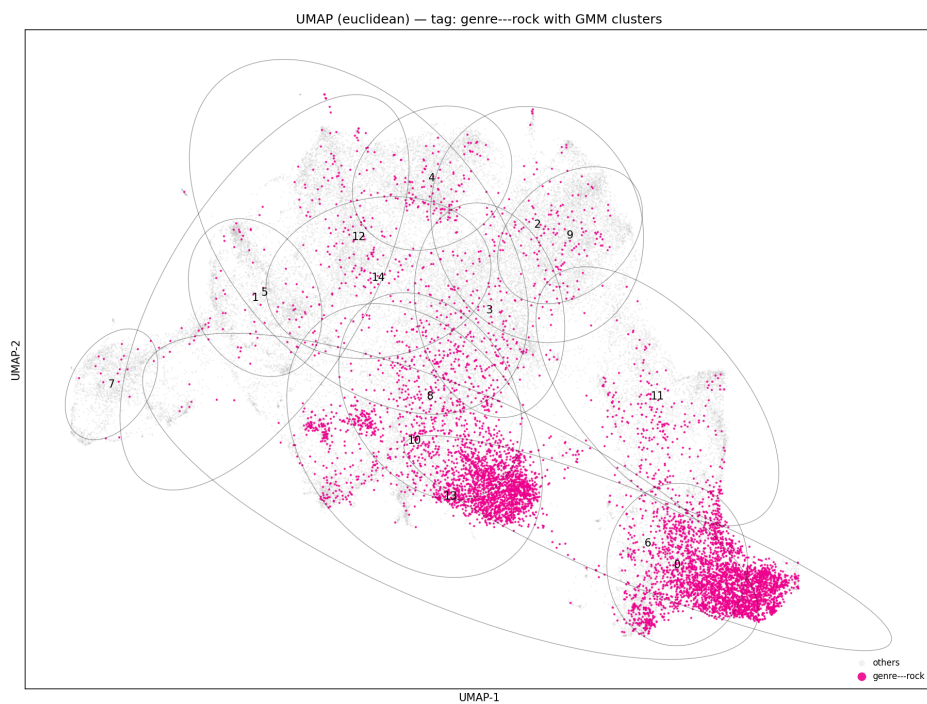


Figure 9: Tag overlay: 'genre-rock' (localized)

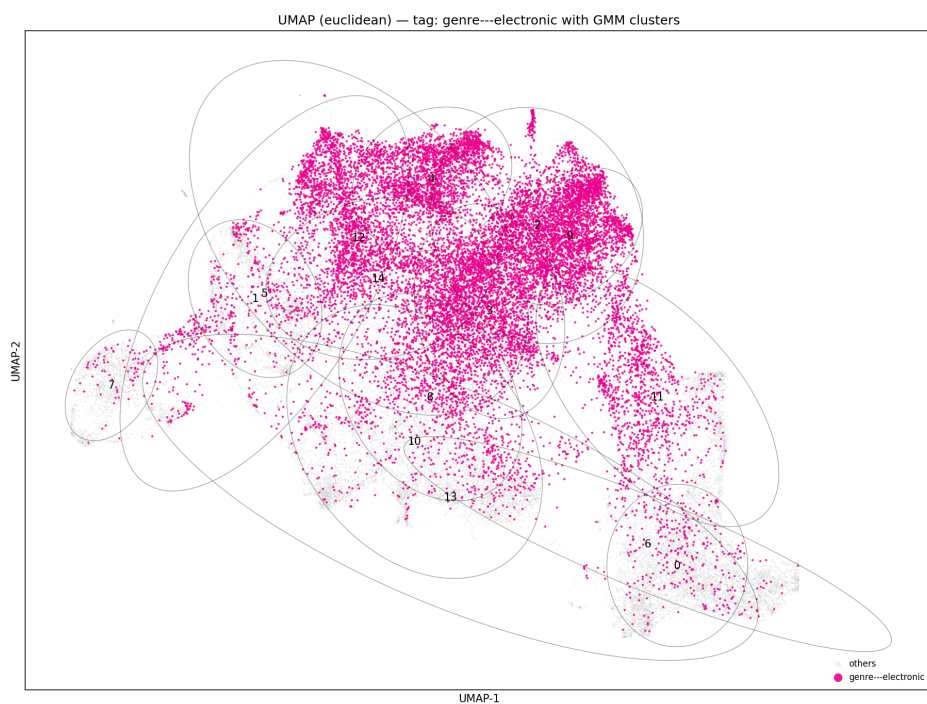


Figure 10: Tag overlay: 'genre-electronic' (dispersed)

4.3. Quantifying Tag Polysemy

We quantify tag polysemy by examining how each tag distributes over the learned components. Importantly, our implementation uses hard labels: for each track, we take a single assigned component and compute $p(c | t)$ from our tag counts (Section 3.8). For each tag t , we compute entropy $H(t)$ (higher means more dispersed), normalized entropy $H_{\text{norm}}(t)$ (scaled by the number of components the tag appears in), top-1 mass (fraction of tracks in the single most common component), and top-3 mass (fraction of tracks in the three most common components).

Table 1: Most polysemous (dispersed) tags

	Tag (t)	n_{tracks}	n_{comp}	$H(t)$	$H_{\text{norm}}(t)$	Top-1 mass	Top-3 mass
1	mood/theme—melodic	1213	15	2.5298	0.9575	0.1146	0.3174
2	mood/theme—dream	951	15	2.5654	0.9473	0.1430	0.3575
3	mood/theme—sad	749	15	2.5384	0.9379	0.1353	0.3698
4	genre—psychedelic	590	15	2.5253	0.9361	0.1271	0.3491
5	mood/theme—melancholic	441	15	2.5317	0.9349	0.1514	0.3583

The entropy ranking in Table 1 (sorted by $H_{\text{norm}}(t)$ and top-1 mass) shows several mood/theme tags among the most dispersed. In particular, ‘mood/theme-melodic’ is the highest-entropy tag in the table, with a normalized entropy of .9575 and top-1 mass of .1146 spanning all 15 components. This indicates that tracks labeled ‘melodic’ do not correspond to a single acoustic mode in embedding space; instead, they spread broadly across components. Other high-entropy tags include ‘mood/theme-dream,’ ‘mood/theme-sad,’ ‘genre-psychedelic,’ and ‘mood/theme-melancholic,’ which similarly exhibit low top-1 mass and substantial entropy across components. Note the descending trend in normalized entropy and ascending trend in top-1 mass, signifying tags in order of most polysemous. The opposite trend is present in Table 2 below in order of most monosemous.

At the other extreme, several genre tags are relatively concentrated, as seen in Table 2. For example, ‘genre-hardrock’ has a normalized entropy of .442 and a top-1 mass of .7122 spanning 12 components. These values suggest that some tags correspond to a more consistent acoustic

Table 2: Most monosemous (localized) tags

	Tag (t)	n_{tracks}	n_{comp}	$H(t)$	$H_{\text{norm}}(t)$	Top-1 mass	Top-3 mass
1	genre—hardrock	490	12	1.0994	0.4420	0.7122	0.8755
2	genre—rap	983	15	1.2739	0.4704	0.6468	0.8691
3	genre—punkrock	729	13	1.2335	0.4801	0.6806	0.8738
4	genre—singersongwriter	776	15	1.3726	0.5089	0.5631	0.8647
5	genre—grunge	375	11	1.2229	0.5099	0.6021	0.9040

Most dispersed example				Most localized example			
Tag (t)		mood/theme—melodic		Tag (t)		genre—hardrock	
n_{tracks}		1213		n_{tracks}		490	
n_{comp}		15		n_{comp}		12	
Entropy $H(t)$		2.5929		Entropy $H(t)$		1.0994	
Normalized entropy $H_{\text{norm}}(t)$		0.9575		Normalized entropy $H_{\text{norm}}(t)$		0.4420	
Top-1 mass		0.1146		Top-1 mass		0.7122	
Top-3 mass		0.3174		Top-3 mass		0.8755	

Table 3: Example of a highly dispersed tag versus a highly localized tag under hard GMM labels.

signature under the embedding model, while others behave as umbrella labels spanning multiple distinct modes.

It should be noted that while these tags are more localized than the polysemous tags in Table 1, they still span most (if not all) of the components. This may imply acoustic similarities not accounted for by the tags (e.g., energy, tempo, and danceability).

4.4. Component Distribution Histograms

To connect the dispersion statistics to a comprehensible picture, we also plot histograms for the most and least polysemous tags for each tag category in each component. In our notebook outputs, these tags are:

Mood/theme: mood/theme-melodic, mood/theme-trailer

Genre: genre-psychedelic, genre-hardrock

Instrument: instrument-synthesizer, instrument-ukulele For each of these tags, we plot the fraction of tracks per component within that tag (GMM components on the x-axis, fraction on the y-axis), as seen in Figure 11. For example, about 65% of songs with the ‘mood/theme-trailer’ tag are in Component 2. To account for tags that have less volume, we normalized the counts to fractions for a



Figure 11: Histograms of component distribution of the most and least polysemous tags.

comparable view of tag distribution. ‘genre-psychedelic’ shows a relatively even distribution across components while ‘genre-hardrock’ favors Component 7. These histograms provide a direct view of whether a tag is dominated by one component (localized) or spread across many (dispersed), and they generally agree with the entropy/top-1 trends: tags that appear “everywhere” in the histograms also rank as high-entropy and low-top-1.

5. Summary

This paper asked whether human-interpretable music tags align with coherent regions of audio embedding space, or whether they systematically span multiple distinct acoustic modes (tag polysemy). Using CLAP audio embeddings for 55,609 MTG-Jamendo tracks, we reduced dimensionality with PCA (44D; 85.39% CEV) and modeled global structure with a Gaussian mixture model. Model selection (AIC/BIC trends) strongly favored full-covariance mixtures, and a frugal $K = 15$ full-covariance GMM yielded overlapping regions when visualized with UMAP. We then operationalized tag polysemy by measuring how each tag distributes across GMM components. The results reveal a clear spectrum: several mood/theme tags (e.g., ‘mood/theme-melodic’) are highly dispersed (high normalized entropy, low top-1 mass), while multiple genre tags (e.g., ‘genre-hardrock’) are localized. Overall, tag ambiguity appears to be structured instead of just annotation noise: many

tags behave like umbrella labels that map to multiple sonic regions in the audio representation.

5.1. Limitations

Our dispersion statistics were computed using hard component assignments, which discards uncertainty in the GMM responsibilities and can overstate separation for borderline tracks. The notion of “acoustic regions” is also model-dependent: the results may vary with the embedding model (CLAP checkpoint), the PCA dimensionality, and the selected K . Additionally, UMAP is used only for visualization and can distort global geometry, so visual “islands” should not be over-interpreted as true distances. Lastly, MTG-Jamendo reflects a particular distribution of open-license music and a fixed tag vocabulary, so tag meaning, generality, and co-occurrence patterns may differ in commercial music data or other cultural contexts.

5.2. Future Work

A direct extension is to define polysemy using *soft* assignments, replacing counts with expected component membership under responsibilities, and to test robustness across reasonable choices of K and PCA dimension. More broadly, comparing multiple embedding spaces (e.g., other music/audio encoders and text-audio models) would help provide more contextualized support. To improve interpretability, future work could “name” components by summarizing characteristic audio traits or by retrieving representative tracks per component. Finally, these analyses could inform MIR systems by motivating new tag models and by refining evaluation metrics to account for polysemous labels.

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